

AYSAN ESMAELY

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RESEARCH INTERESTS

Travel behavior analysis and modeling · Transportation equity and accessibility · Spatial analysis and GIS · Transportation and public health · Machine learning and GeoAI in transportation · Human factors in transportation systems

EDUCATION

M.Sc. in Community and Regional Planning

2024 – December 2025

University of Nebraska–Lincoln

Thesis: *Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska*

Advisor: Dr. Abigail Cochran | GPA: 3.9/4.0

B.Sc. in Urbanism Engineering

2017

Payam Noor University

RESEARCH EXPERIENCE & PROJECTS

Graduate Research Assistant

University of Nebraska–Lincoln, 2024–2025

Transportation Barriers to Vision Care for the Visually Impaired | MATC/ U.S. DOT

- Investigated behavioral and systemic transportation barriers affecting individuals with low vision; analyzed trip-making patterns, mode constraints, and healthcare access limitations.
- Designed and conducted in-depth interviews with healthcare providers; applied qualitative coding to identify travel behavior patterns and unmet mobility needs.
- Contributed to peer-reviewed manuscript preparation; findings published in *Findings* journal (2025).

Telehealth & Transportation Barriers in Rural Nebraska | SENDD / Infrastructure Investment and Jobs Act

- Independently designed a mixed-methods study examining how transportation constraints influence healthcare access and telehealth adoption in rural populations.
- Collected and analyzed data on travel behavior, mode availability, and trip substitution patterns among rural residents and healthcare providers.

Southeast Nebraska Infrastructure Inventory | SENDD / Nebraska Dept. of Economic Development

- Co-designed and deployed a regional infrastructure survey across 15 counties; coordinated data collection from municipal operators to assess transportation and utility system conditions.
- Produced spatial analysis and GIS-based mapping of infrastructure gaps and accessibility deficiencies.

Studio Project

University of Nebraska–Lincoln, MCRP Studio, 2024–2025

City of Ogallala Comprehensive Plan Update |

- Led transportation analysis section: collected and analyzed travel data, produced GIS mapping of road networks and accessibility gaps, and developed transportation recommendations.
- Conducted origin-destination surveys, stakeholder interviews, and public engagement to inform transportation demand analysis.

Thesis Research

University of Nebraska–Lincoln

2024 – 2025

"Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska"

- Conducted in-depth interviews with transportation specialists and planners across Nebraska to investigate institutional responses to mobility barriers.
- Applied systematic qualitative analysis to identify behavioral, policy, and infrastructure-level determinants of transportation disadvantage; thesis published on UNL Digital Commons (2025).

PROFESSIONAL EXPERIENCE

Community Planner, Southeast Nebraska Development District (SEND D)

January 2026 – Present

Lincoln, Nebraska

- Produced GIS maps and spatial analyses for NEPA environmental review, including transportation network delineation, project area mapping, and proximity-to-airport assessments using ArcGIS.
- Supported transportation infrastructure project scoping and environmental review under NEPA compliance.
- Authored and submitted the ECCBG grant for the City of Fairbury (~\$100,000), managing the full grant writing and submission process.
- Assisted communities with transportation components of capital improvement plans and comprehensive plan updates.

Community Outreach Coordinator (Intern), SENDD

Summer 2025

Southeast Nebraska Development District

- Researched telehealth adoption patterns and their relationship to transportation barriers and trip reduction in rural Nebraska.
- Contributed transportation analysis to the Capital Improvement Plan for the City of Waverly.
- Analyzed transportation accessibility conditions and developed recommendations for a Comprehensive Plan update for Pawnee City.

Planner Facilitator, Slum Upgrading Program

2017 – 2020

Gorgan, Golestan Province, Iran

- Led GIS-based mapping to update and formalize city boundary lines and transportation network extents, incorporating informal settlement areas using ArcGIS.
- Built and maintained a comprehensive GIS database of transportation and infrastructure conditions in informal settlements.
- Conducted community engagement to identify mobility challenges and prioritize transportation infrastructure improvements.

Flood Disaster Volunteer

2019

Agh Ghala, Golestan Province, Iran

- Assessed critical transportation infrastructure damage and mapped impacted road networks and access routes.
- Projected mobility needs of vulnerable populations to guide emergency response and aid distribution logistics.

PUBLICATIONS

Published

Nelson-Forcade, C., **Esmaily, A.**, Shepherd, J.D., & Cochran, A.L. (2025). Clinicians' perspectives on transportation barriers to low vision care in the rural Midwest. *Findings*. <https://doi.org/10.32866/001c.129899>

Esmaily, A. (2025). *Transportation Practitioners' Reflections on Addressing Transportation Barriers for Vulnerable Populations in Nebraska*. M.Sc. Thesis. University of Nebraska–Lincoln. digitalcommons.unl.edu

Under Review

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care among individuals with visual impairments in Nebraska. *Under review*.

In Preparation

Esmaily, A., & Cochran, A.L. Transportation practitioners' reflections on addressing transportation barriers for vulnerable populations in Nebraska.

Esmaily, A., Cochran, A.L., Bliss, T., & Borchers-Williams, O. The role of telehealth in overcoming transportation barriers in accessing healthcare in Nebraska.

Esmaily, A. AI agents in qualitative research: Examining machine-generated thematic analysis vs. human coding.

CONFERENCE PRESENTATIONS

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care among individuals with visual impairments in Nebraska. Mar. 2025
Nebraska APA Chapter Conference

Esmaily, A., Nelson-Forcade, C., Shepherd, J.D., & Cochran, A.L. Addressing transportation barriers to low vision care. Nov. 2024
ACSP Annual Conference, Seattle, WA

Cochran, A.L., **Esmaily, A.**, Nelson-Forcade, C., & Choi, I. Understanding and addressing transportation barriers to low vision care in Nebraska. Oct. 2024
UNL Transportation Engineering Seminar Series, Lincoln, NE

TEACHING EXPERIENCE

Teaching Assistant — Urban Planning Workshop

2021 – 2022

Mazandaran University

- Assisted students with GIS-based spatial analysis projects in an urban planning workshop for undergraduate Urbanism Engineering students.

GIS Instructor

2021

Gorgan, Iran

- Taught applied GIS skills including spatial analysis, network mapping, and cartographic visualization to professional practitioners.

TECHNICAL SKILLS

GIS & Spatial Analysis	ArcGIS (proficient), spatial analysis, network analysis, boundary delineation, GIS database management
Statistical & Quantitative	SPSS, survey design and analysis (Lasso, SurveyMonkey), data visualization
Qualitative Methods	In-depth interviewing, thematic coding, Dedoose, mixed-methods research design
Currently Learning	Python, R, machine learning, spatial modeling, GeoAI, travel demand modeling
Other	AutoCAD, grant writing, NEPA compliance, community engagement, stakeholder facilitation

AWARDS & HONORS

Student Scholarship Award for Outstanding Achievement — Nebraska APA Chapter	2025
Student Travel Award — American Planning Association (NPC, Denver)	2025
Graduate Student Travel Award — TRB Annual Conference, Washington D.C.	2025
Nebraska Transportation Center Award — TRB Annual Conference	2024
Graduate Student Travel Award — ACSP Annual Conference, Seattle	2024

PROFESSIONAL MEMBERSHIPS

American Planning Association (APA)
Association of Collegiate Schools of Planning (ACSP)
WTS: Advancing Women in Transportation
Student Planning Association of Nebraska (SPAN)

LANGUAGES

Turkmen: Native **Farsi:** Native **Turkish:** Fluent **English:** Advanced

REFERENCES

Dr. Abigail Cochran

Assistant Professor
Community & Regional Planning
University of Nebraska–Lincoln

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Thesis Advisor & Research Supervisor

Dr. Jaime Lopez

Assistant Professor
Community & Regional Planning
University of Nebraska–Lincoln

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Faculty, UNL

Tom Bliss

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Supervisor, Community Planning